

SIDDHARTH HIRAOU

📍 Buffalo, NY 📞 7167509361 ✉ sidhiraou@gmail.com [LinkedIn](#) [GitHub](#) 🌐 Website

Education

University at Buffalo, Buffalo, New York

August 2024 – January 2026

Masters in Data Science

GPA: 3.6/4.0

Coursework: Data Intensive Computing, Machine Learning, Statistical Learning, Data Models & Query Language, Database Fundamentals

International Institute of Information Technology, Pune, India

June 2020 – June 2024

Bachelor of Engineering in Information Technology

GPA: 4.0/4.0

Coursework: Data Structures & Algorithms, Big Data Analytics, Cloud Computing, Object-Oriented Programming, Artificial Intelligence

Technical Skills And Certifications

Programming Languages: Python, Java, Javascript, C/C++, R, Java, SQL, Bash Shell Scripting

Database: Oracle SQL, MySQL, MongoDB, PostgreSQL, Firebase, Hadoop

Framework/Libraries: Seaborn, Numpy, Pandas, Matplotlib, Tensorflow, Keras, Pytorch, Bigquery, Talend, Alteryx, Kubernetes

Cloud: Amazon Web Services (AWS), Google Cloud Platform (GCP), Oracle Cloud, MongoDB Atlas

Dev Tools: Git, Advanced Excel, Tableau, Power BI, Snowflake, Matlab, AWS, GCP, Azure DevOps, Airflow, Kafka, Apache Spark

Certifications: AWS Certified AI Practitioner, Google Cloud Certified Cloud Practitioner

Experience

Machine Monitoring Systems (MMS)

Sep 2025 – Dec 2025

Data Science Intern

Buffalo, New York

- Developed an end-to-end **fault prediction pipeline** for industrial rotor machines using high-frequency accelerometer data (5–10 kHz), improving fault-pattern separability by 90%.
- Implemented advanced **signal processing (FFT/STFT)** and noise-reduction workflows, boosting model stability and reducing variance by 45%.
- Trained and deployed **2D-CNN/ResNet models** achieving 98.4% accuracy and 98.1% F1-score for early fault detection.

Intisca Technologies

May 2025 – Oct 2025

Data Pipeline Engineer Intern

Dallas, Texas

- Designed and developed an **ETL pipeline** integrating **multiple LLMs (OpenAI GPT-4.0, HuggingFace)** to process user-defined natural language inputs into structured data workflows.
- Conducted detailed financial scoping to estimate operational costs across **AWS infrastructure (EC2, S3, Lambda)** and LLM API usage, enabling informed budgeting and deployment strategy.
- Implemented and orchestrated the pipeline using **Apache Airflow and Python**, ensuring modularity, scalability, and fault tolerance for production-ready deployment

Hanker Analytics

May 2023 – Jul 2023

Data Engineering Intern

Pleasanton, California

- Engineered **Data Quality Management (DQM) model** in Python, leading **36.9%** improvement in analysis report accuracy.
- Implemented an **ML model** using Random Forest to predict causes of disease, with an accuracy of **86.1%**.
- Developed **interactive Power BI dashboard** to visualize client application metrics, driving **300%** increased client engagement.
- Designed **statistical visualization charts** to track company's economic progress, enhancing decision-making & engagement.

IsquareIT

Jun 2022 – May 2023

Research Assistant

Pune, India

- Created a **real-time poultry monitoring system** for Venky's Chicken farm using **ML algo (RCNN)**, achieving model accuracy of **93.1%**.
- Utilized **Python, OpenCV, and CNN-based pattern recognition** to monitor flock behavior, analyzing over **41,000 images** of chicks for health trend identification and risk mitigation
- Conducted **on-ground research** on chicken herd pattern monitoring at IPMT (Institute of Poultry Management and Technology) and implemented attendance monitoring system for farm workers, enhancing workforce accountability and operational efficiency.
- Co-authored the following **research paper** published in April 2023 in the **International Journal of Technological Exploration and Learning (IJTE)**.

Projects

Flock Vision | AWS Lambda, Roboflow, CNN, Git, DynamoDB, Python, PyTorch, IoT

- Designed a smart poultry farm management system using AI-driven flock behavior monitoring and disease detection to optimize productivity for Venky's Chicken, a leading poultry producer.
- Leveraged Python, OpenCV, and CNNs integrated via Roboflow for real-time group pattern recognition, enabling early health alerts and trend analysis.
- Deployed IoT-based temperature regulation and implemented a cloud-synced attendance tracking system to improve worker accountability and maintain optimal farm conditions.

AutoETL | GPT-4 API, LangChain, Airflow, Python, Pandas, Streamlit, Docker

- Built a natural language-driven ETL pipeline generator that converts user prompts (e.g., "Group by region and average sales") into executable Pandas or PySpark code using GPT-4 via LangChain.
- Integrated with Apache Airflow to auto-generate and schedule DAGs, enabling seamless deployment of repeatable data workflows with zero manual coding.
- Provided an interactive Streamlit interface for users to preview transformations, validate data, and export results to CSV or cloud storage, accelerating ETL prototyping and collaboration between engineers and analysts.